

PEARL-Neuro EEG Classification

Predicting Genetic Alzheimer’s-Risk Group from Resting-State EEG

Final Report — Version 2 (consolidated)

Dataset: OpenNeuro ds004796 — 79 genotyped, EEG-imaged subjects

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Headline. Resting-state EEG did not distinguish the genetic Alzheimer’s-risk groups in this cohort. This is a *bounded* null, not an inconclusive one: the same features that find nothing for genotype successfully detect a real subject-level trait (sex) on the identical cohort and analysis protocol. The measurement works; the signal is not there.

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1 What was asked, what was delivered

What was asked. Build a classifier that predicts genetic Alzheimer’s-risk group (derived from APOE ϵ 4 and PICALM genotype) from EEG, using a pitch-synchronous waveform-shape feature family (PSR/PSWT) compared against a conventional spectral baseline, benchmarked against a published result of AUC $\approx$ 0.58.

What was delivered. A validated, reproducible, pre-registered EEG analysis pipeline, and the bounded null result it produced — together with a quantified statement of exactly how large an effect this dataset was capable of detecting. No usable predictor was delivered, because none exists to deliver: the target is not detectable in this cohort at this sample size, and that conclusion is now supported by direct evidence rather than assumed from absence.

2 The headline, in plain language

Think of the pipeline as a metal detector and the dataset as a field. Sweeping the field found no gold. That alone is ambiguous — it could mean the field is empty, or that the detector is broken.

We buried a known coin and confirmed the detector beeps. Using the identical features, subjects, preprocessing and statistics, the pipeline reliably detects a subject’s sex (a trait EEG genuinely encodes) while detecting nothing at all for genotype group.

The field really is empty. The genotype null is a statement about genotype, not about the sensitivity of the measurement.

3 The three acts

3.1 Act I — the original null

The pre-registered primary analysis returned **AUC 0.474** (95% CI [0.266, 0.702], permutation $p = 0.577$) on $n = 64$ subjects: no better than chance.

The pipeline was built so that a positive result would have been believable — label-blind preprocessing enforced by tests, a cross-validation and inference protocol frozen before any label was joined, and provenance tracking from raw file hash to final number.

3.2 Act II — the near-miss, and its retraction

Reviewing the pipeline surfaced two genuine preprocessing defects: the average EEG reference was computed *before* bad-channel interpolation (letting one bad channel contaminate every other channel), and the ICA component count was fixed rather than scaled per subject. Both were fixed on their own merits.

Re-running the *identical* frozen analysis on the corrected data flipped the verdict:

	Original	After the two fixes
AUC	0.474	0.651
95% CI	[0.266, 0.702]	[0.353, 0.764]
Permutation p	0.577	0.041
Verdict	null	<i>apparently</i> positive

This looked like a real finding. It exceeded the published 0.58 benchmark and cleared conventional significance. **It was then subjected to three independent checks and did not survive any of them:**

- **Stability.** Resampling the same 64 subjects put the median AUC at **0.565** — essentially on the covariates-only baseline (0.557) — with **24%** of resamples falling below chance entirely.

- **Attribution.** Isolating each preprocessing fix in a separate full re-run showed *neither* reproduces significance alone: reference-order fix 0.583 ($p = 0.234$), ICA fix 0.602 ($p = 0.174$). Only the specific combination, on this specific sample, crossed $p < 0.05$.
- **Reproducibility.** The 0.651 figure regenerated exactly (0.6509, $p = 0.0410$) at full precision, confirming it was computed correctly — but correctness of computation is not evidence of a real effect.

The result was retracted. It is reported here in full because concealing it would misrepresent what is known.

3.3 Act III — bounding the null

One ambiguity remained. An early diagnostic had shown the PSWT feature family failing a basic sanity check: it could not detect a subject’s *sex* (AUC 0.465 against a pre-declared 0.65 floor), even though resting EEG normally encodes sex. So “we found nothing” might have meant “these features discard all between-subject information.”

Four representations already computed from the corrected data were tested against the sex control — no new analysis of the genotype target was run:

Table 1: Sex control, $n = 64$, full precision (1000 permutations / 2000 bootstrap).

Representation	Cols	AUC	95% CI	perm p	Verdict
baseline_rest	9	0.688	[0.431, 0.816]	0.019	PASS
baseline_rest_msit	16	0.696	[0.421, 0.826]	0.007	PASS
baseline_all	23	0.652	[0.373, 0.819]	0.029	PASS
pswt	18	0.465	[0.272, 0.700]	0.619	FAIL

Three of four decode sex significantly; only the pitch-synchronous family fails. The decisive comparison holds the representation fixed and changes only the target — the same nine columns:

Target	AUC	p	Outcome
Sex (control)	0.688	0.019	signal detected
Genotype (primary)	0.506	0.482	nothing

Matched on features, cohort, preprocessing and protocol. **The null is bounded.**

4 Why the retracted result is the strongest evidence of rigour here

This deserves stating plainly, because it is easy to read Act II as a setback.

A team that ran the corrected preprocessing, obtained AUC 0.651 at $p = 0.041$, and stopped there would have reported a finding that beat the published benchmark. It would have been published, or acted upon, or built into a product. **And it would have been false** — as the resampling and attribution checks subsequently demonstrated.

The value of this engagement is not the answer it produced. It is that the process was built to kill its own most attractive result when that result failed validation, and did so. A pipeline that only ever confirms is worthless; this one demonstrably does not.

5 The detection ceiling — a finding about the dataset

A formal power analysis ($n = 34$ vs 30 , $\alpha = 0.05$ two-sided) quantifies what this cohort could ever have resolved:

Quantity	Value
Minimum detectable AUC at 80% power	0.693
Minimum detectable AUC at 90% power	0.720
Power to detect the published benchmark (0.58)	20%
Power to detect AUC 0.65	57%

Two consequences worth carrying forward:

- **All three passing sex controls sit at or near that 0.693 floor.** Sex is among the most robustly encoded subject-level traits in resting EEG, and this cohort can only just resolve it. A subtler effect has no headroom.
- **At 20% power against the benchmark effect**, a study of this size would fail to detect that effect four times in five *even if it were real and exactly as large as published*.
- **Age is unusable as a control in this cohort** — it spans only 50–63 years, and published EEG brain-age error (≈ 7 years) exceeds that entire range.

These are findings *about the dataset* and are useful to anyone else attempting this question on it.

6 What was actually built

The deliverable is a validated, reproducible EEG analysis pipeline and the bounded null it produced — not a predictor.

Concretely: BIDS ingestion → label-blind preprocessing (including the two fixes, now permanent defaults) → PSR/PSWT and spectral feature extraction → a pre-registered cross-validation, permutation and bootstrap evaluation harness → provenance-traced reporting.

Properties that make it reusable rather than disposable:

- **Label blindness enforced by tests**, not convention — the preprocessing and feature packages have no import path to the genotype labels, asserted by dedicated tests.
- **Pre-registration.** The analysis plan, CV scheme, model class and stopping rules were frozen before any label was joined; changes are append-only amendments.
- **Provenance.** Every derivative records the source file hash, the processing spec, a run ID and the git commit that produced it.
- **289 passing tests**, including real-data regression checks.

A future, adequately powered cohort would be run through this pipeline unchanged.

7 What this does and does not license

This report does *not* say that APOE/PICALM status is invisible in EEG, or that the question is closed in general.

It says that these representations, on this cohort, at this sample size, did not detect it — and it quantifies how large an effect would have had to be to show up ($AUC \geq 0.693$ for 80% power). An effect smaller than that could be entirely real and would have been invisible here.

The distinction matters for anyone deciding whether to pursue this line further: the evidence bounds the effect, it does not exclude it.

8 What a definitive answer would require

1. **A new, adequately powered cohort**, sized from the power analysis on record rather than from availability. This is the binding constraint; no methodological change substitutes for it.
2. **Independent replication data.** ds004796 appears to be the only public EEG dataset genotyped for APOE/PICALM at this scale, so replication requires new primary collection.
3. **Task-state EEG**, if the published literature’s claim that task outperforms rest for this question is to be tested properly — this project’s own task-based benchmark reproduction landed at chance and the gap remains characterised but unresolved.

9 Limitations and known defects

9.1 Limitations

- **Hard sample ceiling.** 79 subjects have both EEG and genotype; all were used. 64 survive QC for the rest-based primary analysis.
- **No independent replication cohort exists** (see above).
- **Benchmark mismatch.** The published 0.58 came from task-state (MSIT) EEG with a substantially richer feature pipeline on a different cohort; it is a characterised methodological difference, not a validated comparison point in either direction.
- **Cohort sensitivity.** A change of only 5 subjects (a QC side effect) moved the point estimate by 0.177 AUC — inherent to this sample size.
- **The bound applies to representations tested.** Three independent representations carry subject-level information and find no genotype effect; this does not prove no representation could.

9.2 Known defects annex

Recorded rather than concealed:

- **Report-writer overwrite defect (fixed).** Report writers used fixed output paths with no overwrite protection. This silently destroyed 131 lines of the pre-registered analysis plan (two appended amendments) during re-runs; recovered from version control. Report writes are now guarded — the frozen plan is write-once and all other reports archive any prior version before overwriting.

- **Hardcoded narrative (fixed).** Two report generators contained prose describing a specific expected result, including a stale “verdict (frozen: NULL, AUC 0.474)” claim that would have been emitted unchanged alongside a different result. Narrative is now generated from computed values or not at all, enforced by test.
- **Duplicated disclaimer (fixed).** The inference module served by the container carried its own copy of the output disclaimer, so strengthening the primary definition would have left the shipped surface unchanged. Now a single imported definition.
- **Control-figure provenance (corrected).** An earlier report draft described the sex control — a *failed positive* control — as a passed negative control, and did not state which preprocessing each control figure came from. Corrected: 0.849 / 0.465 / 0.490 are from the corrected preprocessing; 0.830 / 0.486 / 0.459 from the original.

10 Deliverables index

Artefact	Path
This report	reports/final_report_v2.tex
Bounded null (Phase 6)	reports/phase6_bounded_null.md
Detection ceiling	reports/phase6_power_analysis.txt
Retraction of the 0.651 result	reports/phase5_stage3_validation.md
Stop decision (no further analysis)	reports/phase7_stop_decision.md
Delivery decision (what ships)	reports/phase7_delivery_decision.md
Pre-registered analysis plan	reports/analysis_plan_frozen.md
Original Phase 3 model (frozen)	data/derivatives/models_phase3to4_frozen/
Reference model artefact + card	data/derivatives/models_stage3/
Reproduction instructions	RUNBOOK.md, INSTALL.md

This project answered its question. The answer was “no”. The reason that answer is worth having is that the pipeline which produced it was built so a “yes” would have been believable — pre-registered, label-blind, provenance-traced, and willing to discard its own best result when that result failed validation.